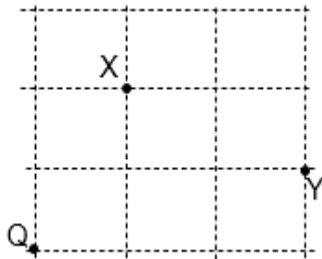


- 19** An isolated point charge is placed at point Q. The electric potential at point X is measured to be 720 V.

The relative positions of the three points Q, X, and Y are shown on the grid of squares below.



If the electric potential at X is 720 V, what is the electric potential at Y?

A	509 V	B	581 V	C	720 V	D	936 V
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Answer: A

Electric potential due to a point charge follows:

$$V \propto 1/r$$

Distance from Q to X:

$$r_X = \sqrt{1^2 + 2^2} = \sqrt{5} = 2.24$$

Distance from Q to Y:

$$r_Y = \sqrt{3^2 + 1^2} = \sqrt{10} = 3.16$$

$$\frac{V_Y}{V_X} = \frac{r_X}{r_Y} = \frac{2.24}{3.16} = 0.707$$

$$V_Y = 720 \times 0.707 = 509 \text{ V}$$